

ARIA — Use Cases

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Feature: Monocular Depth Perception (No Dedicated RGB-D Sensor)

ARIA recovers the Z-coordinate through monocular depth estimation (Depth-Anything or MiDaS) combined with camera and table-height calibration geometry, replacing a \$200-600 RGB-D camera with a \$10-30 webcam or ESP32-CAM.

- **Electronics Contract Manufacturer:** standardizing 40 sorting stations on monocular depth instead of RGB-D removes roughly \$16,000 in sensor cost across the line, with savings scaling directly with every added station.
- **Retail Chain Backroom Automation:** a 200-store rollout of automated returns sorting becomes financially realistic only once the per-station sensor cost is removed, since RGB-D pricing at that scale would otherwise dominate the capital budget.
- **Agricultural Cooperative with Multiple Packing Sheds:** several remote packing sheds with limited capital budgets can each afford a sorting arm once the RGB-D line item is eliminated, where a single RGB-D-equipped arm might have been the only affordable option.
- **Franchise Quick-Service Kitchen Automation:** a franchise chain testing light kitchen-prep automation across many small outlets keeps per-outlet hardware cost low enough to justify a chain-wide pilot, which an RGB-D-dependent design would price out.
- **Automotive Aftermarket Rework Station:** a small rework line correcting mis-assembled trim parts uses monocular depth to justify automating several low-volume stations that a single RGB-D-equipped cell could never have covered on the same budget.

Feature: Object Detection and Segmentation (Vision Agent)

YOLO handles fast per-frame detection while SAM2 provides pixel-accurate segmentation masks, giving the Vision Agent its first pass at what is in view even under partial occlusion.

- **Cluttered Bin-Picking in Small-Parts Warehousing:** a warehouse sorting station handling densely packed mixed-part bins benefits from a detection pipeline validated under configurable occlusion conditions, ensuring reliable performance before deployment on genuinely cluttered bins.
- **Pharmaceutical Blister-Pack Line:** blister packs partially hidden behind cartons on a crowded conveyor are still segmented reliably, letting a packaging inspector flag mis-fills without redesigning the line layout for a clear line of sight.

Feature: Low-Cost 5-DoF Actuation (SG90 / MG995 Servos)

The arm uses lightweight SG90 servos for fine joints and load-bearing MG995 servos for heavier joints, giving a full manipulator for a fraction of the cost of an industrial-grade arm.

- **Small Produce Packing Facility:** hobby-servo-class arms deployed across five packing lines cost less in total than a single industrial-grade arm, while still meeting the force requirements of handling fruit and packaged goods.
- **Independent Hardware Repair Shop:** a shop repairing consumer electronics uses a low-cost arm to hold and reposition small housings during repair, a task where industrial-grade force and precision would be unnecessary overkill.

- **Rural Clinic Supply Room:** a low-budget clinic automates light supply-bin organization with a low-cost arm, a use case that would never be funded at industrial-arm pricing.
- **Community Recycling Sorting Station:** a municipal recycling facility with a tight equipment budget uses several low-cost arms to pre-sort lightweight recyclables, where a single industrial arm's cost would have limited the facility to one sorting line instead of several.
- **School and University STEM Lab Fleets:** a department can equip an entire lab section with one arm per two or three students instead of sharing a single industrial cell, turning manipulation coursework from a demonstration into hands-on practice.

Feature: Affordance Agent (Use-Case-Aware Grasp Selection)

Grasp candidates from GraspNet or AnyGrasp are filtered by an Affordance Agent that selects the grasp matching how an object is actually used, a cup by its handle, a tool by its shaft, rather than the nearest reachable surface.

- **E-Commerce Fulfillment Lane:** a single picking station handles mixed small-goods SKUs by re-ranking grasp candidates per item in real time, avoiding gripper retooling between product categories.
- **Hand-Tool Distribution Warehouse:** tools of varying shapes (screwdrivers, wrenches, pliers) are picked by their functional grip point rather than an arbitrary surface, reducing drops caused by unstable generic grasps.
- **Kitchenware Assembly Line:** mugs, cups, and handled containers are consistently grasped by handle rather than by body, reducing scuffing and breakage compared to a fixed suction or parallel-jaw default.
- **Hardware Store Restocking:** mixed fasteners, tools, and packaged goods on a single restocking cart are each grasped appropriately without a human needing to pre-sort them by shape beforehand.
- **Precision Agriculture Grasp Strategies:** selective harvesting, seedling transplanting, and soil sampling are delicate tasks where material-aware, gentle grasp strategies are directly applicable, with the system serving as a testbed before field deployment.
- **Greenhouse Plant Handling:** pot repositioning and pruning tasks rely on affordance learning to distinguish a fragile stem grasp from a robust pot grasp.
- **Pharmacy and Laboratory Container Handling:** precise, repeatable manipulation of vials, petri dishes, and syringes benefits from affordance-based grasp selection, with planned material recognition allowing fragile glass to be handled differently from plastic.
- **Automotive Interior Trim Assembly:** clips, fasteners, and wiring harness connectors that each demand a different grip point are handled without a separate fixture per part type, useful on low-volume interior sub-assembly lines.

Feature: Confidence-Gated Decision-Making

Every action carries a confidence score, and when confidence drops below a threshold, the system stops and asks the user before proceeding instead of guessing through uncertainty.

- **Pharmacy Compounding Lab:** an ambiguous grasp on an irregularly shaped reagent bottle triggers a pause and operator confirmation rather than a low-confidence guess, reducing spillage and breakage risk.
- **Museum Artifact-Handling Pilot:** a conservation team piloting automated handling of replica artifacts relies on confidence gating to avoid the arm ever acting on an uncertain grasp near a fragile object.
- **Semiconductor Cleanroom Handling:** components that are difficult to distinguish visually under cleanroom lighting trigger a confirmation step rather than a wrong-part pick, protecting a high per-unit cost inventory.
- **Jewelry or Small Precious-Goods Handling:** a workshop automating light handling of valuable small items uses confidence gating as a built-in checkpoint before any uncertain grasp, reducing the risk of loss or damage to high-value stock.

- **Nuclear Fuel Handling Simulation:** a facility modeling remote fuel-assembly handling uses confidence gating as a proxy for the kind of mandatory human sign-off that real regulatory frameworks require before any uncertain manipulation step.

Feature: World Model Agent (Semantic Map with Lifecycle Tracking)

The World Model Agent maintains a running map of known objects, including coordinates, color and material attributes, and a lifecycle state that moves through detected, tracked, lost, recovered, moved, and removed.

- **Retail Backroom Inventory Automation:** the semantic map tracks which items have moved to which bin across a shift, and flags a missing item to a floor associate instead of silently assuming a detection error.
- **Tool-Crib Management in a Machine Shop:** checked-out tools are tracked through a lifecycle of removed and returned states, giving the shop a live picture of what is currently on the floor versus in the crib.
- **Library or Archive Materials Handling:** physical media or archival boxes moved between shelving locations are tracked by the world model, reducing the labor of manual inventory audits.
- **Hospital Supply Cabinet Tracking:** consumable medical supplies moved in and out of a stock cabinet are tracked through the same lifecycle states, supporting more accurate reorder timing than periodic manual counts.
- **Cold-Chain Food Storage:** trays moved between staging and freezer zones are tracked through the same lifecycle states, helping a facility flag items left too long outside temperature-controlled storage.

Feature: Health Monitoring (Servo Temperature, Drift, Calibration Drift)

The system continuously watches servo temperature, mechanical drift, calibration drift, camera frame rate, and inference latency as it operates.

- **Double-Shift Manufacturing Line:** a servo trending toward overheating is flagged hours before failure, allowing a scheduled swap during a shift changeover instead of an unplanned line stoppage.
- **Unattended Warehouse Operation:** calibration drift accumulating over a multi-day unattended run is caught and flagged before it degrades pick accuracy enough to cause a cascade of failed picks.
- **Remote or Field-Deployed Cell:** a facility with infrequent on-site technician visits uses health telemetry to schedule a single maintenance visit that addresses several developing issues at once, rather than reactive call-outs.
- **Multi-Unit Fleet Deployment:** a company operating several arms across different sites uses aggregated health data to prioritize which unit needs maintenance first, rather than servicing units on a fixed schedule regardless of actual condition.
- **Offshore or Remote Energy Facility:** an installation with limited technician access uses drift and temperature trends to plan the one maintenance trip that a helicopter or boat transfer allows per rotation, rather than discovering a failed servo on arrival.

Feature: Learning Agent (Self-Supervised Skill Improvement)

Logged operational data, pick success rate, IK error, recovery rate, feeds a Learning Agent that gradually improves a fixed library of skills without requiring a full retraining cycle.

- **Contract Assembler Onboarding a New Part:** pick success on a newly introduced connector type improves over several days of normal operation without an engineer manually retraining the cell.

- **Seasonal Produce Packer:** grasp performance on a new seasonal fruit variety improves automatically across the first days of the season, rather than requiring a pre-season engineering pass for every new crop.
- **Multi-Site Franchise Deployment:** performance improvements learned at one site's most common product mix can inform tuning at other sites facing similar product profiles, reducing redundant engineering effort.
- **Returns Processing Center:** grasp handling of a growing variety of returned product types improves over time as the volume and diversity of returns naturally expands the logged experience the Learning Agent draws on.
- **Automotive Aftermarket Parts Distributor:** as the SKU catalog grows with new part numbers each quarter, grasp success on newly stocked items improves through ordinary operation instead of a dedicated retraining pass per catalog update.

Feature: Evaluation Agent (Structured Failure Classification)

Failures are automatically sorted into categories, missed object, object slipped, IK failure, collision, or perception error, giving troubleshooting a defined starting point.

- **Food Processing Sorting Line:** an uptick in dropped items is traced to the object-slipped category for one specific product shape, pointing directly at a grasp-selection fix rather than a hardware fault.
- **Small Parts Manufacturing Cell:** recurring IK-failure classifications on one arm posture reveal a workspace layout issue rather than a solver bug, redirecting the fix from software to floor-plan adjustment.
- **Multi-Shift Facility with Rotating Technicians:** a shared failure-classification log lets a new shift's technician pick up troubleshooting where the previous shift left off, without re-diagnosing from scratch.
- **Equipment Vendor Support Contract:** a vendor providing remote support across many customer installations uses the categorized failure logs to triage which sites need an on-site visit versus a remote software fix.
- **Electronics Rework and Repair Depot:** a depot handling many different device models uses the failure taxonomy to tell whether a rising defect rate on one model traces to perception error versus a genuinely tighter mechanical tolerance.

Feature: LLM-Based Planning Agent with Multi-Agent Orchestration

A local LLM-based Planning Agent uses function-calling to break commands into goals, subgoals, and actions, and hands work off to the right specialist agent among roughly fifteen cooperating agents.

- **High-Mix, Low-Volume Manufacturer:** a line is retasked for a new short-run product using plain-language instructions instead of a full reprogramming cycle, letting a small operations team retask the cell within the same day.
- **Contract Packaging House Serving Multiple Clients:** switching between different clients' packaging specifications during the same shift is handled through updated natural-language instructions rather than separate fixed programs per client.
- **Small Job-Shop with No Dedicated Robotics Engineer:** floor supervisors without programming backgrounds can issue new task instructions directly, reducing dependence on scarce robotics programming talent.
- **Pop-Up or Temporary Production Line:** a manufacturer setting up a short-term production run for a seasonal or promotional product configures the cell through spoken instructions rather than a full engineering setup, matching the temporary nature of the line.
- **Textile Cut-and-Sew Facility:** a garment line that changes product style several times a week is reconfigured through spoken instructions instead of a dedicated program per style, matching the pace of fast-fashion production runs.

Feature: Chain-of-Thought Explainability and Logging

Every task is accompanied by an explicit, plain-language log of the reasoning behind each step, from locating the object through to confirming task completion.

- **Clinical Diagnostics Lab:** the logged decision trail for every automated sample-handling batch is retained alongside the batch record, giving auditors a timestamped explanation of what the arm did and why.
- **Aerospace Component Handling:** a quality assurance team uses the reasoning log as supporting documentation when a handled component is later flagged for inspection, showing exactly how it was picked, moved, and placed.
- **Insurance or Liability Review After an Incident:** in the event of a dropped or damaged item, the logged chain of reasoning gives a factual record of the arm's decision process rather than relying on inference from video alone.
- **Third-Party Compliance Audit:** an external auditor reviewing a facility's automation processes uses the logged reasoning trail as documentary evidence of consistent, rule-following behavior over a sampled period.
- **Food Safety (HACCP) Compliance Recordkeeping:** a food handling line retains the reasoning log as part of its critical-control-point documentation, giving inspectors a step-by-step account of how a product was moved and handled.

Feature: Safety Agent (Collision and Cable Awareness)

Motion planning accounts for both workspace obstacles and the arm's own trailing cables, with a dedicated Safety Agent checking for collisions before execution.

- **Shared Human-Robot Bench in Small Assembly:** cable-aware, obstacle-aware planning keeps the arm's own wiring and the shared workspace clear of collisions without a full safety cage, lowering integration cost near people.
- **Cluttered Small-Lab Bench:** a bench crowded with equipment and cabling from other instruments is navigated safely because the planner treats both fixed obstacles and its own cable slack as constraints.
- **Mobile or Reconfigurable Cell:** an arm redeployed to a new bench layout each week relies on the Safety Agent to re-validate collision safety automatically rather than requiring a manual safety re-certification each time.
- **Shared Workshop or Makerspace:** a community workshop where several people work near the arm at different times benefits from continuous collision checking rather than a one-time safety review at installation.
- **Automotive Paint Shop Cell:** an arm working near spray booths and overhead hose reels uses the same cable- and obstacle-aware planning to avoid contact with equipment that a rigid, pre-programmed path would not adapt to.

Feature: Graceful Degradation and Search Behavior

When something is missing or a task fails, the system actively searches the workspace, looks behind obstacles, and attempts recovery rather than simply stopping.

- **Unattended Overnight Warehouse Restocking:** an item that rolls out of its expected bin position is searched for and recovered automatically, letting the task queue continue through the night without technician intervention.
- **Remote Agricultural Sorting Shed:** intermittent connectivity or supervision means a stalled task queue could sit idle for hours; automatic search-and-recovery keeps throughput up between infrequent check-ins.
- **Multi-Tasking Small Manufacturing Cell:** a temporarily misplaced part during a busy shift is recovered without halting the broader task queue, preserving overall line throughput despite a single missed pick.

- **Weekend or Holiday Skeleton-Crew Operation:** a facility running with minimal staffing over a weekend relies on the arm's own recovery behavior to handle minor disruptions that would otherwise require paging in a technician.
- **Disaster-Relief Supply Depot:** a temporary depot sorting donated supplies with a skeleton crew relies on the same search-and-recovery behaviour to keep processing moving when items are stacked unpredictably.

Feature: Fine-Grained Skill Library (Slide, Roll, In-Hand Manipulation)

Beyond basic pick and place, the skill library includes slide, roll, sweep, and in-hand manipulation actions such as rotating a held object to a target orientation.

- **Precision Fastener Assembly:** a screwdriver is rotated to a specific orientation before insertion into a fastener, avoiding the need for a separate fixture or a human hand-off step.
- **Small Parts Inspection Station:** components are rolled in-hand to present different faces to a camera for multi-angle visual inspection, without needing a dedicated rotating fixture.
- **Packaging Line Requiring Sweep or Slide Actions:** loose small items are swept or slid into position on a tray ahead of a pick step, a motion not well served by pick-and-place alone.
- **Cable or Connector Assembly Bench:** a connector is rotated in-hand to align its keying feature before mating with a socket, a task that requires fine in-hand reorientation rather than a simple approach-and-place motion.
- **Electronics Kitting Station:** small connectors and headers are reoriented in-hand before being placed into a kit tray in a consistent orientation, reducing downstream errors on the assembly line that receives the kits.

Feature: Tracking Agent (Real-Time Moving-Object Tracking)

A dedicated Tracking Agent follows objects that are moving rather than static, with real-time reacquisition and motion prediction.

- **Slow-Moving Conveyor Packaging Line:** the Tracking Agent predicts an item's position as it moves, allowing correctly timed grasps without requiring the line to stop for every pick.
- **Vibration-Prone Workbench:** parts that shift slightly during a nearby operation are reacquired by the Tracking Agent rather than treated as lost, avoiding a false missing-object failure.
- **Shared Workspace with Human-Moved Items:** when a coworker repositions an item mid-task, the Tracking Agent reacquires it rather than requiring the whole task to restart.
- **Sorting Bin Feed from an Upstream Process:** items arriving continuously from an upstream chute or feeder are tracked as they settle into position, letting the arm begin planning a grasp before the item comes fully to rest.
- **High-Speed Bottling Line Sampling:** a quality-control arm pulls periodic bottle samples off a moving line for inspection, using motion prediction to time the grasp without requiring the line to pause.

Feature: Teleoperation Learn Mode

A separate learn mode allows a person to teleoperate the arm directly, logging images, joint states, and commands for later use in improving autonomous behavior.

- **New-Part Onboarding at a Small Manufacturer:** an experienced technician teleoperates the arm through several successful handling cycles on an unfamiliar part, and the logged demonstrations bootstrap autonomous performance.

- **Cross-Site Skill Transfer:** demonstrations recorded at one facility for a specific product can be reused to bootstrap a second facility handling the same product, shortening that site's ramp-up period.
- **Difficult Edge-Case Capture:** rare but recurring failure scenarios are demonstrated directly by a technician so the Learning Agent has example data to train against, rather than waiting for the case to occur autonomously.
- **Retiring-Employee Knowledge Capture:** a long-tenured operator's manual handling technique for a tricky legacy part is captured through teleoperation before that knowledge is otherwise lost when the employee retires.
- **Defense and Hazmat Training Simulation:** an operator's technique for handling a simulated hazardous item is recorded through teleoperation, giving a training program reusable demonstration data instead of relying only on live drills.

Feature: Inverse Kinematics Solver Benchmarking and Trajectory Generation

Multiple IK solver families, analytical, numerical optimisation, neural, and library-based, are benchmarked on the arm's geometry, and quintic or cubic spline trajectory generation converts the result into motion the servos can follow smoothly.

- **Inverse Kinematics Solver Benchmarking:** four solver families are benchmarked on the same set of test poses with identical metrics, letting any new IK method plug into the benchmark interface for a direct, apples-to-apples comparison.
- **High-Speed Pick-and-Place Line Trajectory Tuning:** a manufacturer evaluating jerk-minimized trajectories against a legacy trapezoidal motion profile uses the benchmarking interface to select a smoother motion profile before committing to production hardware.
- **Robotics Startup Hardware Selection:** a startup choosing between servo and IK-solver combinations for a new low-cost arm design uses the benchmarking interface to make an evidence-based choice before committing to a bill of materials.

Overall Use Cases

- **Manufacturing Automation Feasibility Studies:** before purchasing industrial hardware, a company can prototype a manipulation task using the task planner, skill library, and world model together, producing realistic success-rate estimates for a pick-place or sort-and-stack workflow.
- **Warehouse Micro-Logistics Simulation:** small-parcel sorting, label reading and placement, and bin picking are all handled with the existing skill library, with the world model tracking object positions to mimic a real warehouse's dynamic inventory.
- **Quality Control Inspection Planning:** the inspect skill, active perception, and multi-view object reconstruction combine to answer how many views the system needs to reliably classify a defect, before any physical inspection station is built.
- **Human-Robot Collaboration Workflow Design:** designing shared workspaces requires knowing when the robot stops to ask for input and when it proceeds autonomously; tuning the confidence threshold together with the world model lets a workflow designer set this without risk to human operators.
- **Robot Application Cost Estimation:** a manager can measure task completion time and success rate for a proposed application and estimate ROI before purchasing hardware, using planning time, success rate, and recovery rate together as the basis for that estimate.
- **Environmental Sample Collection Planning:** designing a robot to collect soil or water samples in the field requires planning reach envelope, grasping strategy, and failure recovery together, all testable before deployment in a remote environment.

- **EOD (Explosive Ordnance Disposal) Robot Simulation:** remote manipulation planning in hazardous environments relies on the confidence-gated approval workflow together with the teleoperation learn mode, which records operator strategies for later autonomous improvement.
- **Hazardous Material Handling Simulation:** designing remote manipulation workflows for nuclear, chemical, or biological material handling draws on the world model, semantic map, and planning pipeline together, ahead of any hardware deployment.
- **Search and Rescue Planning:** the search behaviour, systematic scan, looking behind obstacles, querying a remote operator, combines with moving-target tracking to model both locating a stationary object and following a moving one, such as a survivor.
- **Mining and Extractives Remote Handling Simulation:** sample and core-tray handling in a hazardous pit or underground environment can be modeled end-to-end, combining the world model, safety agent, and confidence gating before any hardware is exposed to a live site.
- **Insurance and Risk Underwriting for Automation Deployments:** an insurer evaluating the risk profile of a proposed automation deployment can use the logged failure taxonomy, recovery rate, and confidence-gating behaviour as objective inputs into a policy, rather than relying on a vendor's marketing claims alone.
- **Cold-Chain and Pharmaceutical Logistics Feasibility Study:** a distributor evaluating automated handling of temperature-sensitive goods can combine the world model's lifecycle tracking with the affordance and confidence-gating agents to estimate feasibility and risk before any refrigerated hardware pilot.